Polysomnography

Behrouz Jafari, MD[a], Vahid Mohsenin, MD[a,b,*]

KEYWORDS

- Polysomnography • Rapid eye movement
- Sleep disorders • Sleep study

The science of sleep and the specialty of sleep medicine have evolved rapidly since the initial attempts in the 1930s to develop a consistent framework to describe the complexity of sleep. The methods initially used to characterize correlates of sleep entailed the recording of brain electrical activity in animals in 1875[1] and the subsequent demonstration of the ability to detect and characterize wakeful activity in humans in 1929.[2] Detection and recording of human heart electrical activity were developing at about the same time[3] with identification of cardiac electrical waveforms by Einthoven[4] in 1895. In 1909, Cajal first delineated a network-intermingled collection of nerves and fibers (reticular formation) extending from the spinal cord to the thalamus.[5] Brainstem lesions high up at the midbrain produce continuous electroencephalogram (EEG) characteristics of sleep. The first continuous overnight EEG sleep recordings in humans were published in 1937.[6] The tracings on miles of paper recorded with an 8-ft–long drum polygraph were summarized using a data reduction scheme called sleep staging (stages A, B, C, D, and E), with stages A and B approximately corresponding to the current stage N1, stage C corresponding to stage N2, and stages D and E corresponding to stage N3. They recognized phenomena, such as the fragmentation and fallout of alpha rhythm sleep spindles and high-amplitude slow waves.

The combination of breathing and brain monitoring in physiologic recordings to identify pathologic conditions during sleep evolved in the mid-twentieth century.[7] Later, additional parameters were added when limb myoclonus was described in 1953.[8] Rapid eye movement (REM)–associated respiratory and cardiac effects were identified in 1953 by Aserinsky and Kleitman[9] and later more formally incorporated into the stages of REM sleep. In 1957, Dement and Kleitman[10] proposed the first classification based on an understanding that REM and non-REM (NREM) sleep alternate in successive cycles during the night. They suggested 4 stages (1–4), with stage 1 corresponding to stage N1 at the start of the night and stage R toward morning, stage 2 corresponding to stage N2, and stages 3 and 4 to stage N3.

Establishing a standard for describing sleep recording technique and sleep stage scoring varied widely from one laboratory to the next. Different terminology was used from one sleep center to the next; for example, REM sleep might be called D sleep, paradoxic sleep, desynchronized sleep, or even unorthodox sleep. In response to this circumstance, a committee was formed by members of the Sleep Research Society to standardize the data acquisition and scoring of sleep.[11] The committee recommended the recording of at least 1 EEG derivation, 2 electrooculogram (EOG) derivations, and 1 submental electromyogram (EMG). The recommendation required that sleep be scored in arbitrary epochs of 20 to 30 seconds with a single stage assigned to each epoch. They divided sleep into 5 stages: stages 1 through 4 of NREM sleep and stage REM sleep. Despite occasional suggestions for modifying the system,[11–16] the Rechtschaffen and Kales (R and K) manual remained the standard staging system for human sleep studies for almost 4 decades until 2004, when the board of directors of the American Academy of Sleep Medicine

[a] Section of Pulmonary, Critical Care and Sleep Medicine, Yale Center for Sleep Medicine, Yale University School of Medicine, New Haven, CT, USA
[b] Department of Medicine, Yale University School of Medicine, Yale Center for Sleep Medicine, New Haven, CT, USA
* Corresponding author. The John B. Pierce Laboratory, 290 Congress Avenue, New Haven, CT 06519.
E-mail address: Vahid.mohsenin@yale.edu

Clin Chest Med 31 (2010) 287–297
doi:10.1016/j.ccm.2010.02.005

(AASM) commissioned the development of a new manual for the scoring of sleep, including not only sleep staging but also rules for scoring other parameters (ie, arousals, respiratory, cardiac, and movement events).[17] The *AASM Manual for the Scoring of Sleep and Associated Events*, covering all aspects of sleep scoring, were published in 2007.[18] The AASM requires that the new scoring rules be followed in AASM-accredited sleep centers and laboratories.

TECHNIQUES

The polysomnography (PSG) uses various methods to simultaneously and continuously record neurophysiologic, cardiopulmonary, and other physiologic parameters over the course of several hours, usually during an entire night (overnight PSG). PSG provides information on the physiologic changes occurring in many different organ systems in relation to sleep stages and wakefulness. It allows qualitative and quantitative documentation of abnormalities of sleep and wakefulness, sleep-wake transition, and physiologic function of other organ systems that are influenced by sleep. Many of these, such as sleep apnea, may not be present during wakefulness.

Four types of sleep studies are available, depending on the number of physiologic variables recorded[19]:

- Level I. Standard PSG includes EEG (frontal, central, and occipital derivations), EOG, chin EMG, ECG, and recordings of airflow, respiratory effort, oxygen saturation, and limb EMG.[18] A technician is in constant attendance.
- Level II. Comprehensive portable PSG studies are essentially the same as level I, except that a heart rate monitor can replace the ECG and a technician is not in constant attendance.
- Level III. Modified portable sleep apnea testing is a cardiorespiratory study that includes ventilation (at least 2 channels of respiratory movement or respiratory movement and airflow), heart rate or ECG, and oxygen saturation. Ventilation in this case is measured with at least 2 channels of respiratory movement or of airflow. Personnel are needed for preparation, but the ability to intervene is not required for all studies.
- Level IV. Continuous (single or dual) bioparameter recordings where devices that measure a minimum of 1 parameter, usually oxygen saturation, are used.

Video monitoring, although not required, is extremely valuable, from diagnostic and medicolegal perspectives. Additional variables, such as 16-channel EEG recording for seizures, end-tidal CO_2 monitoring, pulse transit time, and esophageal pressure monitoring, can be added according to the clinical indications. The AASM recommends performing at least 6 hours of overnight recording.[19] The neurophysiologic, ECG, and sound channels are sampled at 100 to 1000 Hz with 12- to 20-bit resolution, respiratory mechanical channels at 50 Hz, and pulse oximetry and body position, at a significantly lower rate. The modern-day computerized PSG systems provide powerful tools for technicians and physicians to customize data acquisition. Reference electrodes, resolution, and sensitivity can be changed and digital filters added or removed to minimize recording artifacts. Many of these functions can be performed even after the data are acquired, which allows interpretation of the studies despite malfunction of certain equipment during the recording.[20]

The application of electrodes and sensors to a patient is the most important part of a sleep study. If done poorly, the quality of the data is compromised and a significant amount of the night is spent troubleshooting and problem solving. Filters should not be used to compensate for poor-quality electrode application, because changing filter settings can significantly alter the data, which affects how the study is analyzed.

Electroencephalography

EEG is the recording of surface electrical activity of the brain. Only limited EEG data are obtained during a PSG recording to help identify stages of sleep and wakefulness.

Reliable EEG recording begins with accurate measurement of the human skull, as dictated by the International Federation of Societies for Electroencephalography and Clinical Neurophysiology 10–20 system of electrode placement (**Fig. 1**).[21] Each of the points on the 10–20 map indicates a possible electrode site. Each electrode site is designated with a letter or letters and a number. The letters FP, F, C, P, and O represent frontal pole, frontal, central, parietal, and occipital, respectively. M represents the mastoid process. Odd numbers are used to denote the left-sided electrode placements; even numbers are used to denote the right-sided electrode placements. Z denotes midline electrode placement sites.

It is well recognized that alpha activity associated with wakefulness usually is most prominent when recorded from the occipital region. Although

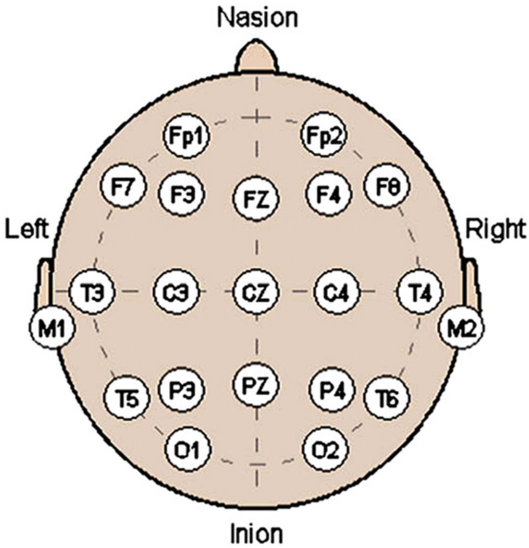

Fig. 1. Schematic representation of the 10–20 electrode placement system.

Williams and coworkers[22] argued as early as 1973 for including a bipolar occipital lead as a matter of routine, it was not adopted as standard technique until 1985 for use on the multiple sleep latency test[23] and 1992 for central nervous system arousal scoring on overnight PSGs.[24] Typically, a monopolar mastoid-referenced derivation from O1, O2, O3, or O4 is suggested.

The recommended derivations are F4-M1, C4-M1, and O2-M1.[18] A minimum of 3 EEG derivations is required to sample activity from the frontal, central, and occipital regions (M1 and M2 refer to the left and right mastoid processes).[18] This recommendation is partly based on K complexes maximally represented with frontal lobe electrodes, sleep spindles maximally seen with central electrodes, and delta activity maximally seen with frontal electrodes.[25] An accepted alternative is to use FZ-CZ, CZ-OZ, and C4-M1 with backup electrodes at FPZ, C3, O1, and M2 for alternative electrode displays in case of primary electrode malfunction.[18]

Electro-oculography

An EOG records changes that occur in the corneoretinal potential with eye movements during sleep and wakefulness. The cornea and retina form a dipole, with the cornea positive in relation to the retina. A movement in the eyes changes the electrical signal in the EOG electrodes, which is recorded as a deflection. The recommended EOG derivations are E1 (left eye)–M2 and E2 (right eye)–M2. E1 is placed 1 cm below the left outer canthus. E2 is placed 1 cm above the right outer canthus for ease of recognition of REMs.[18]

The EOG is important in the evaluation of sleep and sleep stages. REMs are one of the signs of REM sleep and are essential for the scoring of REM sleep. These movements are seen as sharp bursts of electrical activity. At the time of sleep onset, slow eye movement can be seen.

Electromyogram

Submental and leg (tibialis anterior) EMG recordings are performed routinely during PSG. The submental EMG recordings are essential for scoring sleep stages, especially REM sleep. Three electrodes are placed for submental EMG recordings, to have a backup in case one of them malfunctions during the sleep study. Typically, the submental EMG tone is lowest during REM sleep. It is also helpful in detecting sleep bruxism. Bilateral leg EMG recordings are used to diagnose periodic limb movements of sleep. Additional EMG recordings can be used under special circumstances. For example, both upper and lower extremity EMGs can be recorded for suspected REM behavior disorder, and gastrocnemius muscle EMG could be recorded for diagnosing nocturnal leg cramps.[20]

Air Flow Measurement

The gold standard for measurement of air flow is body plethysmography and pneumotachography; however, they are considered unsuitable for routine PSG.[26] Currently airflow is measured with the help of a thermistor or with a nasal pressure monitor.

The thermistor measures changes in the electrical conductance in response to temperature changes in the probe, which occur with inspiration and expiration. Currently, use of an oronasal thermal sensor is recommended for detection of apnea and is used in conjunction with a nasal pressure transducer to allow the detection of hypopneas.[18] The ability of the nasal pressure transducer to adequately detect respiratory events is comparable to the gold standard approach of the pneumotachometer.[27,28]

Respiratory Effort Measurement

Different monitors have been used in the past to evaluate respiratory effort, including strain gauges, piezoelectric transducers, and impedance pneumography. The current recommendation is esophageal manometry or inductance plethysmography for monitoring and recording respiratory effort.[18] Esophageal manometry involves the passage of a thin flexible tube containing a pressure transducer through the nose or mouth, into the esophagus. Diaphragmatic contraction during inspiration causes a drop in thoracic pressure that is transmitted to the esophagus and detected by the pressure transducer. The reverse occurs during expiration. Esophageal pressure measurements are a sensitive and qualitative index of inspiratory effort. Because esophageal manometry is not well accepted or tolerated by most subjects undergoing PSG, inductance plethysmography has become the method of choice in most sleep laboratories.

Respiratory inductance plethysmography measures respiratory effort from body surface movement. Inductance is the property of a circuit or circuit element that opposes a change in current flow. Bands that contain transducers (which consist of an insulated wire sewn onto an elasticized band in the shape of a horizontally oriented sinusoid) are placed around the rib cage and abdomen.[29] With inspiration and expiration, the changes in the cross-sectional area of the chest and abdomen cause a proportional change in the diameter of the transducer. This change in the diameter of the transducer alters the inductance. These monitors can be calibrated to a known volume, which provides qualitative and quantitative analyses of breath volumes.

Snoring

Snoring is recorded from a snore microphone, and the signal is displayed as a continuous waveform. Technician notes are the most useful way of assessing the severity of snoring. Snoring signal is also prominently reflected in chin EMG and in nasal pressure tracing.

Cardiac Monitoring

Standard recording of cardiac rhythm during PSG is by use of a modified lead II electrograph with 1 lead just inferior to the right clavicle and the other on the left side at the level of the seventh rib. Current recommendations are that several cardiac parameters, such as average heart rate during sleep, highest heart rate during sleep, highest heart rate during recording, bradycardia, and other arrhythmias, be reported in all sleep study interpretations.[18]

Measurement of Oxygenation

The most commonly used noninvasive method for the continuous monitoring of blood oxygen is the use of pulse oximetry, in which the oxygen saturation of arterial blood ($Sa{O_2}$) is determined by the passage of 2 wavelengths of light (650 nm and 805 nm) through a pulsating vascular bed from 1 sensor to another. The light is partially absorbed by the oxygen-carrying molecule, hemoglobin, depending on the percent of the hemoglobin saturated with oxygen. Several factors influence the accuracy and reliability of pulse oximetry, however, including the sensor location and type[30]; the presence of abnormal hemoglobin species, such as carboxyhemoglobin or methemoglobin[29]; reduced skin perfusion caused by hypothermia; hypotension or vasoconstriction[31]; and change in heart rate and circulation time.[32]

Other Monitoring Devices

Body position is also an important piece of information to monitor, because snoring and upper airway obstruction during sleep are influenced by gravity.[33,34] It is common to see a worsening of snoring and sleep apnea when a subject is in the supine position. Continuous recording of body position is determined by a sensor, which is usually placed on the chest. A common problem with body position sensor is that it may not correlate closely with the actual subject's position and therefore it is important for a technologist to record the subject's body position via camera/video monitoring to confirm accuracy of the body position monitor.

STAGING SLEEP

In 2007, the new *AASM Manual for the Scoring of Sleep and Associated Events* was published, providing a comprehensive reference for the evaluation of PSGs.[18] According to the new criteria, the following staging system was proposed:

- Stage W (wakefulness)
- Stage N1 (NREM stage 1 sleep)
- Stage N2 (NREM stage 2 sleep)
- Stage N3 (NREM stage 3 sleep)
- Stage R (REM sleep).

The change in sleep staging differentiates the scoring rules from those of the R and K manual published in 1968. The basic principle of sleep staging is that the night is divided into 30-second sequential periods, known as epochs. Each epoch is assigned a stage. If 2 or more stages can be identified during a single epoch, the stage comprising the greatest portion of the epoch is assigned. Although an assessment of the overall flow of sleep might be better obtained by relying on identifying the start and end of each stage irrespective of epochs, such a scoring method is cumbersome in the presence of highly fragmented sleep, as may be seen in patients with obstructive sleep apnea syndrome (OSA).

Wakefulness (Stage W)

Wakefulness is defined by the presence of 8 to 13 Hz sinusoidal alpha rhythm in the posterior head regions on the EEG (**Fig. 2**)[18] and greater than 50% of an epoch having this rhythm. Alpha appears with eye closure and attenuates with eye opening. About 10% to 20% of normal subjects have little or no alpha rhythm.[35] When alpha rhythm is not clearly discernable by visual inspection of the EEG, wakefulness can still be scored if the EOG demonstrates any 1 of 3 markers of alertness:

- Eye blinks; blinking results in conjugate vertical eye movements at a frequency of 0.5 to 2 Hz, which are visible on the EOG
- Reading eye movements (which consist of a slow phase followed by a rapid movement in the opposite direction)
- Irregular conjugate eye movements with normal or high chin muscle tone.

Stage N1 Sleep

Although the AASM manual recognizes that many signs of drowsiness are discernable before alpha rhythm is lost, it does specify a single time of sleep onset. This is defined as the start of the first epoch scored as any stage other than stage W, recognizing that in most subjects this is stage N1.[18] This stage is defined by the presence of low-amplitude, mixed-frequency activity of predominantly 4 to 7 Hz (**Fig. 3**).[18] The EOG generally shows slow eye movements and the chin EMG is often lower in amplitude than in stage W. The EEG may show vertex sharp waves, which are sharply contoured waves maximal over the central region, distinguishable from the background EEG and lasting less than 0.5 second. None of these additional features is required for scoring stage N1, although their presence may be helpful in equivocal situations. In subjects who do not have alpha rhythm and show low-amplitude, mixed-frequency activity with the eyes closed even in wakefulness, discerning sleep onset can be problematic. In these circumstances, stage N1 sleep should be scored when slowing of background EEG frequencies by greater than or equal to 1 Hz, vertex sharp wave, or slow eye movements is observed.[36] The onset of slow eye movements usually precedes the other changes and is the criterion most likely to be applied. Because slow eye movements are usually seen when alpha rhythm is still present, the sleep latency is likely to be slightly shorter in subjects who do not show alpha rhythm.[36]

Stage N2 Sleep

Stage N2 sleep is defined by the presence of 1 or more K complexes without associated arousals or 1 or more trains of sleep spindles in the first half of the epoch or in the second half of the prior epoch.[18] A K complex is a well-defined biphasic wave easily discernable from the background EEG with total duration greater than or equal to 0.5 seconds, usually maximal frontally (**Fig. 4**). A sleep spindle is a train of waves with a frequency of 11 to 16 Hz, but usually 12 to 14 Hz, with duration greater than or equal to 0.5 second, usually maximal centrally. There are no defined amplitude criteria for a K complex or a sleep spindle. K complexes can be associated with arousals induced by environmental stimuli or by sleep apnea. In some patients with OSA, sleep can be highly fragmented by repeated apnea-induced arousals with frequent K complexes. K complexes induced by arousals do not represent a deeper level of sleep and so cannot be used to designate stage N2 sleep, unless a spontaneous K complex or sleep spindle is also present in the same epoch. K complexes and sleep spindles are intermittent phenomena and may not always be present in successive epochs. Their absence does not indicate that sleep has reverted to stage N1 unless an arousal occurs or a major body movement followed by slow eye movements is present. Stage N2 sleep terminates when there is a transition to stages W, N3, or R.[36]

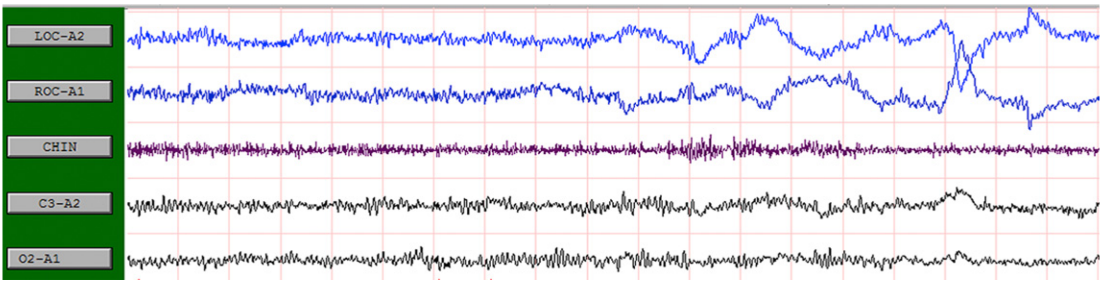

Fig. 2. Wakefulness with presence of alpha waves and REMs.

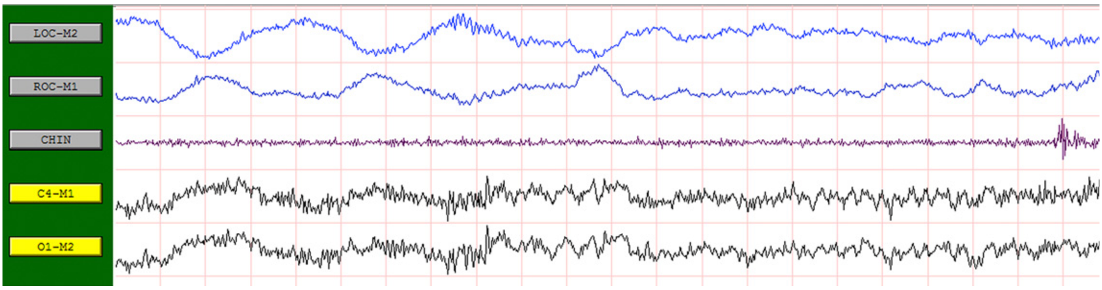

Fig. 3. Stage N1 with mixed-frequency EEG and rolling eye movements.

Stage N3 Sleep

Stage N3 sleep is scored when greater than or equal to 20% of an epoch consists of slow wave activity, defined as waves of frequency of 0.5 to 2 Hz with peak-to-peak amplitude greater than 75 mV recorded over the frontal regions (**Fig. 5**).[18] The frequency band is a subset of the classically defined delta frequency range (<4 Hz) and the term, *delta sleep*, should not be used. Sleep spindles may persist in stage N3 sleep but are not required for scoring.

The neurophysiologic genesis of slow waves differs from that of K complexes,[37] such as growth hormone, is maximally released during slow wave sleep rather than stage N2.[38,39]

Stage R, REM Sleep

REM sleep is scored if an epoch includes low-amplitude, mixed-frequency EEG, REMs, and low chin EMG tone (**Fig. 6**).[18] The EEG is similar to that in stage N1, but some subjects have more prominent alpha frequencies, often at a frequency somewhat slower than that of their alpha rhythm when they are awake. REMs are defined as conjugate, irregular, sharply peaked eye movements with the duration of the initial deflection usually less than 500 milliseconds. Low chin EMG tone implies that the chin EMG amplitude is no higher than in any other stage and is usually at the lowest level of the entire PSG. Certain other phenomena, if present, may support scoring stage R but are not required. These include saw tooth waves (2–6 Hz sharply contoured or triangular, often serrated waves maximal over the central head regions and frequently preceding bursts of REMs) and transient muscle activity, previously known as phasic muscle twitches (burst of EMG activity lasting <0.25 seconds superimposed on low tone background, which may be present in the chin, anterior tibialis, or EEG-EOG derivations). Detailed rules define when a period of stage R sleep ends.[18] In general, once stage R is scored, subsequent epochs should continue to be scored as stage R until there is a definite change to another stage, an arousal or major body movement is followed by slow eye movements, or K complexes or sleep spindles occur in the absence of REMs, even if chin EMG tone remains low. If an epoch contains REMs and the chin EMG tone is low, stage R should be scored even if K complexes or sleep spindles are present. This latter situation frequently occurs during the first REM period of the night when REM and NREM phenomena intermix. There are also rules concerning the scoring of transition epochs between epochs of definite stage N2 and definite REM sleep. In outline, stage R should be scored if chin EMG tone is low and K complexes and sleep spindles are absent, even if REM has not yet commenced. If K complexes or sleep spindles are present in the transition epochs without REMs, however,

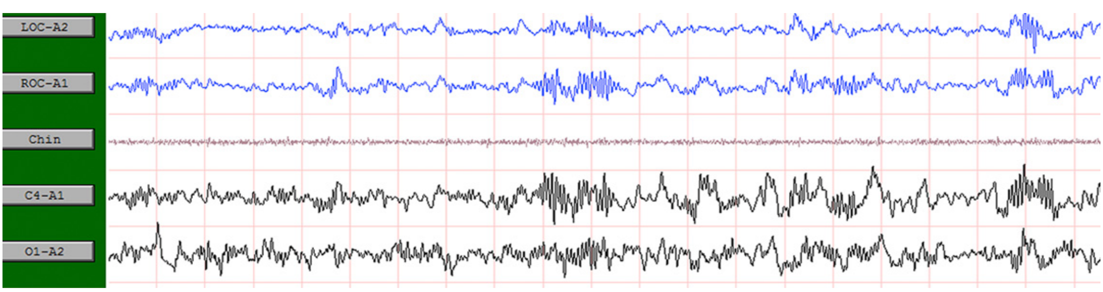

Fig. 4. Stage N2 with K complex and sleep spindles.

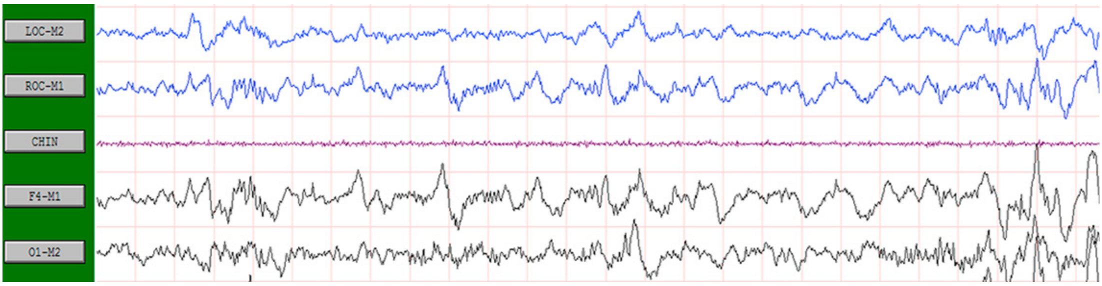

Fig. 5. Stage N3 with delta waves.

these should be scored as stage N2, even if chin muscle tone is low.[18,36]

Major Body Movement

The term, *movement time*, was used in the R and K manual[40] to classify epochs when more than 50% of the EEG was obscured by body movements and muscle artifact. The AASM manual eliminated this term, replacing it with the concept of major body movements. Epochs with major body movements are scored as stage W if any alpha rhythm, even comprising less than half the epoch, is present, or if the preceding or following epoch is scored as stage W. Otherwise, the epoch is given the same stage as the epoch that follows.[18] The logic behind this rule is that major body movements generally result in an arousal or at least a change to a lighter stage of sleep.[36]

Indications for PSG

The Standards of Practice Committee of the AASM reviewed the indications for PSG in 2005 and recommended the following[41]:

- Sleep-related breathing disorders, including OSA, central sleep apnea syndrome (CSA), Cheyne-Stokes respiration (CSR), and alveolar hypoventilation syndrome; upper airway resistance syndrome; the present reference or gold standard for evaluation of sleep and sleep-related breathing is the PSG
- Narcolepsy, parasomnias, sleep-related seizure disorders, restless legs syndrome, periodic limb movement sleep disorder, depression with insomnia
- For continuous positive airway pressure (CPAP) titration in patients with sleep-related breathing disorders; for the assessment of treatment results in some cases; with a multiple sleep latency test in the evaluation of suspected narcolepsy; in evaluating sleep-related behaviors that are violent or otherwise potentially injurious to the patient or others; and in certain atypical or unusual parasomnias. PSG may be indicated in patients with neuromuscular disorders and sleep-related symptoms, to assist in the diagnosis of paroxysmal arousals or other sleep disruptions thought to be seizure related, in a presumed parasomnia or sleep-related seizure disorder that does not respond to conventional therapy, or when there is a strong clinical suspicion of periodic limb movement sleep disorder.

PSG is not routinely indicated to diagnose chronic lung disease; in cases of typical, uncomplicated, and noninjurious parasomnias when the diagnosis is clinically evident; for patients with seizures who have no specific complaints consistent with a sleep disorder; to diagnose or treat restless legs syndrome; for the diagnosis of circadian rhythm sleep disorders; or to establish a diagnosis of depression.

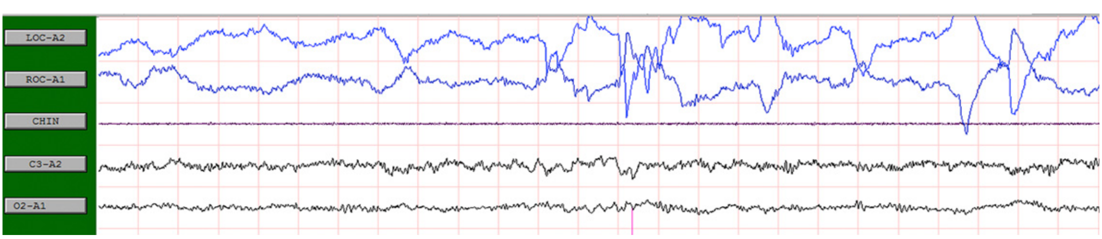

Fig. 6. Stage REM with rapid eye movements and saw tooth waves.

PORTABLE PSG

Comprehensive standard PSG is expensive and often considered inconvenient for patients who have to travel to a center and spend a night away from home. It has been argued that many patients who have sleep-disordered breathing in the form of OSA do not require such a comprehensive procedure to obtain an accurate diagnosis. For this reason, different types of devices have been developed to test for sleep apnea at home or outside the usual environment of a sleep center. Portable monitoring, therefore, in this context means technology that can be performed outside a sleep center and often is unattended. Such devices may be as simple as overnight oximetry or include all the same leads as performed with standard PSG.[42]

In 1994 the AASM noted that the available portable devices had insufficient reliability to allow for widespread usage.[43] In 2005 and 2007, different type of studies emerged, which suggested that in highly selected populations, the outcomes with regards to the use of CPAP were at least equivalent with portable compared with standard PSG.[44,45] Later in 2007, another article was published by the AASM, which set forth some clinical guidelines for use of portable studies.[46] Finally, these devices appeared to have broken through many of the barriers that were preventing it from more widespread usage. AASM divided the technology into different parameters, which may be a more useful way to approach portable devices types (**Tables 1** and **2**).[46] Type 2 devices require experienced technologist to avoid problems with data acquisition that tend to occur with these devices. These types of studies might be best for institutionalized patients who have trouble getting to a sleep center or for hospitalized patients.[42]

Type 3 devices typically do not monitor EEG and sleep stage variables, so they cannot detect arousals from sleep, but data are sufficient to detect most apneas and hypopneas. As a result, hypopneas with arousals or respiratory effort–related arousals are not scored, leading to an underestimation of the degree of sleep-disordered breathing. Likewise, because the apnea-hypopnea index (AHI) from a type 3 device is calculated by dividing the number of apneas and hypopneas (only scored with desaturations) by the entire study duration rather than total sleep time, tends to be underestimated. These calculation changes cause the AHI tends to be derived by a type 3 to be lower than that derived by PSG, which decreases the diagnostic sensitivity.

The information that can be collected from a type 4 device includes frequency of apneas, frequency of hypopneas (with desaturation only), an AHI (as divided by total recording time), baseline Sa_{O_2}, mean Sa_{O_2}, frequency of desaturation, and nadirs of Sa_{O_2}. Therefore, not only do all of the limitations of type 3 devices apply to type 4 devices but also type 4 devices cannot differentiate between obstructive and central/mixed apneas. There are other limitations regarding types 3 and 4 devices. Unlike PSG, in which there are standardized rules for scoring events, there are no published standardized rules for portables. Many devices use proprietary algorithms to score events, and a user cannot alter the scoring.[42] Similarly, the characteristics of overnight pulse oximetry depend on the criteria used to define events. When using a quantitative measure, such as a 4% desaturation, the sensitivity is low for diagnosing OSA; however, the specificity is acceptable.[47–49] When using more qualitative criteria, however, such as analyzing the tracing for frequent desaturations, often known as a saw tooth pattern, the sensitivity improves but the

Table 1
Types of monitoring systems during sleep

	Type 1	Type 2	Type 3	Type 4
Number of leads	≥7	≥7	≥4	1–2
Types of leads	EEG, EOG, EMG, ECG, airflow, effort, oximetry	EEG, EOG, EMG, ECG, airflow, effort, oximetry	Airflow, effort, oximetry, ECG	Oximetry + other (usually airflow)
Setting	Attended usually in a sleep center	Unattended	Unattended	Unattended

Data from Ferber R, Millman R, Coppola M, et al. Portable recording in the assessment of obstructive sleep apnea. ASDA standards of practice. Sleep 1994;17(4):378–92.

Table 2
AASM clinical guidelines: parameters used for portable monitors

Parameter	Examples
Oximetry	
Respiratory monitoring	Effort Airflow Snoring End-tidal CO_2 Esophageal pressure
Cardiac monitoring	Heart rate Heart rate variability Arterial tonometry
Measures of sleep/ wake activity	EEG Actigraphy
Body position	Accelerometer
Other	

Data from Collop NA, Anderson WM, Boehlecke B, et al. Clinical guidelines for the use of unattended portable monitors in the diagnosis of obstructive sleep apnea in adult patients. Portable Monitoring Task Force of the American Academy of Sleep Medicine. J Clin Sleep Med 2007;3(7):737–47.

specificity falls.[50] A more recent study showed poor consistency between pulmonary physicians interpreting overnight oximetry readings.[51] The low level of consistency suggests there are significant problems with either approach that result in increased false-negative or false -positive tests.

Several studies that have used portable and standard PSG in a highly selected patient population have shown no significant differences in the outcome measures between these two types of monitoring. The absence of difference in the outcomes were, in large part, due to employment of highly trained sleep specialists and the training and education of the patients on the use of these devices.[44,45,52]

Based on the AASM recommendations, portable devices should record airflow, respiratory effort, and blood oxygenation. The most accurate airflow signals would include a thermister and a nasal pressure sensor and the most accurate effort signals would use respiratory inductance plethysmography.

Choosing an appropriate population is important when considering use of portable device to diagnose OSA.[53] As discussed previously, most of the studies showing positive outcomes with portables have been done on patients who had a high likelihood of having OSA. In addition, confounding disorders, such as respiratory disease or cardiac disease, are usually a reason to exclude patients for portables testing. The presence of other comorbid sleep disorders, such as insomnia, restless legs syndrome, or parasomnias, are also likely to reduce the accuracy of portables, and their use should be avoided.[42]

SUMMARY AND FUTURE

PSG is an essential tool for diagnosis of variety of sleep disorders. The results of PSG should be interpreted in the context of a patient's history and medications and observation in the sleep laboratory. As new technologies evolve, it is expected that the field also will evolve. Further work is needed to determine if computerized scoring, with or without human revision, may one day reliably replace visual scoring in normal and abnormal sleep. Improved techniques to measure and quantify sleep itself will allow for more meaningful assessment of sleep disruption that can lead to the recognition of new disorders and better predictions of the outcomes of these disorders.

REFERENCES

1. Caton R. The electric current of the brain. Br Med J 1875;2:278.
2. Berger H. Uber das elektroenkelphalogramm des Menshcen [About electroencephalogram of Menshcen]. Arch Psychiatr Nervenkr 1929;87: 527–70 [in German].
3. Walter A. A demonstration on man of electromotive changes accompanying the heart beat. J Physiol 1887;8:229–34.
4. Einthoven W. Uber die form des menschlichen electrocardiogramms [About the form of the human electrocardiographic program]. Arch Gesamte Physiol 1895;60:101–23 [in German].
5. Cajal RS. Histologie du systeme nerveux de l'homme et des vertebres. Paris: Norbert Maloine; 1909.
6. Loomis AL, Harvey N, Hobart GA. Cerebral states during sleep, as studied by human brain potentials. J Exp Psychol 1937;21:127–44.
7. Gastaut H, Tassinari CA, Duron B. Polygraphic study of diurnal and nocturnal (hypnic and respiratory) episodal manifestations of Pickwick syndrome. Rev Neurol (Paris) 1965;112:568–79.
8. Symonds CP. Nocturnal myoclonus. J Neurol Neurosurg Psychiatry 1953;16(3):166–71.
9. Aserinsky E, Kleitman N. Regularly occurring periods of eye motility and concomitant phenomena during sleep. Science 1953;118:273–4.
10. Dement W, Kleitman N. Cyclic variations in EEG during sleep and their relation to eye movements, body motility, and dreaming. Electroencephalogr Clin Neurophysiol 1957;9(4):673–90.

11. Hori T, Sugita Y, Koga E, et al. Proposed supplements and amendments to 'a manual of standardized terminology, techniques and scoring system for sleep stages of human subjects', the Rechtschaffen & Kales (1968) standard. Psychiatry Clin Neurosci 2001;55(3):305–10.
12. Himanen SL, Hasan J. Limitations of Rechtschaffen and Kales. Sleep Med Rev 2000;4(2):149–67.
13. Hirshkowitz M. Standing on the shoulders of giants: the standardized sleep manual after 30 years. Sleep Med Rev 2000;4:169–79.
14. McGregor P, Thorpy MJ, Schmidt-Nowara WW, et al. T-sleep: an improved method for scoring breathing-disordered sleep. Sleep 1992;15(4):359–63.
15. Shepard JW. Atlas of sleep medicine Armonk. New York: Futura Publishing Company; 1991.
16. van Sweden B, Kemp B, Kamphuisen HA, et al. Alternative electrode placement in (automatic) sleep scoring (Fpz-Cz/Pz-Oz versus C4-A1). Sleep 1990; 13(3):279–83.
17. Iber C, Ancoli-Israel S, Chambers M, et al. The new sleep scoring manual: the evidence behind the rules. J Clin Sleep Med 2007;3:107.
18. Iber C, Ancoli-Israel S, Chesson A, et al. The AASM manual for the scoring of sleep and associated events—rules, terminology and technical specifications. 1st edition. Westchester (IL): American Academy of Sleep Medicine; 2007.
19. ASDA Standards of Practice. Portable recording in the assessment of sleep. Sleep 1994;17:378–92.
20. Kakkar RK, Hill GK. Interpretation of the adult polysomnogram. Otolaryngol Clin North Am 2007;40: 713–43.
21. Jasper HH. The ten twenty electrode system of the International Federation. Electroencephalogr Clin Neurophysiol 1958;10:371–5.
22. Williams RL, Karacan I, Hursch CJ. EEG of human sleep: clinical applications. New York: Wiley; 1974.
23. Carskadon MA, Dement WC, Mitler MM, et al. Guidelines for the multiple sleep latency test (MSLT): a standard measure of sleepiness. Sleep 1986;9(4):519–24.
24. EEG arousals. Scoring rules and examples: a preliminary report from the Sleep Disorders Atlas Task Force of the American Sleep Disorders Association. Sleep 1992;15(2):173–84.
25. Silber MH, Ancoli-Israel S, Bonnet MH, et al. The visual scoring of sleep in adults. J Clin Sleep Med 2007;3(2):121–31.
26. Redline S, Budhiraja R, Kapur V, et al. The scoring of respiratory events in sleep: reliability and validity. J Clin Sleep Med 2007;3(2):169–200.
27. Heitman SJ, Atkar RS, Hajduk EA, et al. Validation of nasal pressure for the identification of apneas/hypopneas during sleep. Am J Respir Crit Care Med 2002;166(3):386–91.
28. Thurnheer R, Bloch KE. Monitoring nasal conductance by bilateral nasal cannula pressure transducers. Physiol Meas 2004;25(2):577–84.
29. Kryger MH, Roth T, Dement WC. Principles and practice of sleep medicine. 3rd edition. Philadelphia: Saunders; 2004.
30. West P, George CF, Kryger MH. Dynamic in vivo response characteristics of three oximeters: Hewlett-Packard 47201A, Biox III, and Nellcor N-100. Sleep 1987;10(3):263–71.
31. Berg S, Haight JS, Yap V, et al. Comparison of direct and indirect measurements of respiratory airflow: implications for hypopneas. Sleep 1997; 20(1):60–4.
32. Farre R, Montserrat JM, Ballester E, et al. Importance of the pulse oximeter averaging time when measuring oxygen desaturation in sleep apnea. Sleep 1998;21(4):386–90.
33. McEvoy RD, Sharp DJ, Thornton AT. The effects of posture on obstructive sleep apnea. Am Rev Respir Dis 1986;133(4):662–6.
34. Oksenberg A, Khamaysi I, Silverberg DS, et al. Association of body position with severity of apneic events in patients with severe nonpositional obstructive sleep apnea. Chest 2000;118(4): 1018–24.
35. Santamaria J, Chiappa KH. The EEG of drowsiness in normal adults. J Clin Neurophysiol 1987;4(4):327–82.
36. Silber MH. Staging sleep. Sleep Med Clin 2009;4(3): 343–52.
37. Steriade M, Amzica F. Slow sleep oscillation, rhythmic K-complexes, and their paroxysmal developments. J Sleep Res 1998;7(Suppl 1):30–5.
38. Holl RW, Hartman ML, Veldhuis JD, et al. Thirty-second sampling of plasma growth hormone in man: correlation with sleep stages. J Clin Endocrinol Metab 1991;72(4):854–61.
39. Van Cautier E, Plat L, Copinschi G. Interrelations between sleep and the somatotrophic axis. Sleep 1998;21:553–66.
40. Rechtschaffen A, Kales A. A manual of standardized terminology, techniques and scoring system for sleep stages of human subjects. Washington, DC: US Government Printing Office NIH publication; 1968.
41. Kushida CA, Littner MR, Morgenthaler T, et al. Practice parameters for the indications for polysomnography and related procedures: an update for 2005. Sleep 2005;28(4):499–521.
42. Collop NA. Portable monitoring. Sleep Med Clin 2009;4:3.
43. Ferber R, Millman R, Coppola M, et al. Portable recording in the assessment of obstructive sleep apnea. ASDA standards of practice. Sleep 1994; 17(4):378–92.
44. Mulgrew AT, Fox N, Ayas NT, et al. Diagnosis and initial management of obstructive sleep apnea

without polysomnography: a randomized validation study. Ann Intern Med 2007;146(3):157–66.
45. Whitelaw WA, Brant RF, Flemons WW. Clinical usefulness of home oximetry compared with polysomnography for assessment of sleep apnea. Am J Respir Crit Care Med 2005;171(2):188–93.
46. Collop NA, Anderson WM, Boehlecke B, et al. Clinical guidelines for the use of unattended portable monitors in the diagnosis of obstructive sleep apnea in adult patients. Portable Monitoring Task Force of the American Academy of Sleep Medicine. J Clin Sleep Med 2007;3(7):737–47.
47. Douglas NJ, Thomas S, Jan MA. Clinical value of polysomnography. Lancet 1992;339(8789):347–50.
48. Gyulay S, Olson LG, Hensley MJ, et al. A comparison of clinical assessment and home oximetry in the diagnosis of obstructive sleep apnea. Am Rev Respir Dis 1993;147(1):50–3.
49. Williams AJ, Yu G, Santiago S, et al. Screening for sleep apnea using pulse oximetry and a clinical score. Chest 1991;100(3):631–5.
50. Series F, Marc I, Cormier Y, et al. Utility of nocturnal home oximetry for case finding in patients with suspected sleep apnea hypopnea syndrome. Ann Intern Med 1993;119(6):449–53.
51. Ramsey R, Mehra R, Strohl KP. Variations in physician interpretation of overnight pulse oximetry monitoring. Chest 2007;132(3):852–9.
52. Berry RB, Hill G, Thompson L, et al. Portable monitoring and autotitration versus polysomnography for the diagnosis and treatment of sleep apnea. Sleep 2008;31(10):1423–31.
53. Epstein LJ, Kristo D, Strollo PJ. Clinical guideline for the evaluation, management and long-term care of obstructive sleep apnea in adults. J Clin Sleep Med 2009;5(3):263–76.